Weekly Reflections

Week 14 Updates

During week 14, the following features were implemented:
- Created the admin dashboard page with EJS and added a protected route for it in Express.
- Protected admin dashboard route using isAuthenticated
- Added a one-hour session timeout in the existing session config
- Styled the admin login and dashboard pages with CSS
- Added admin header/navigation for the admin pages
- Created the admin special offer editing form on the dashboard
- Ensures that the marketing page's special offer content updates from admin changes
- Added a public review posting form on the marketing page
- Review submission saves into MongoDB and updates the average rating and review count for the property
- Updated the review calculation
- Added a public contact popup form replacing the old mailto logic. Submissions will post on the admin dashboard and save to MongoDB. Admins can clear the messages through a checkbox-based delete function.
- Added an admin “View Reviews” pop-up on the dashboard
- Did some minor UI fixes on the marketing page, like spacing and placeholder text.
- Ensured there was placeholder text on the review form and contact form, and that the review form only accepts ratings between 1 and 5.
- Tested out the password hashing and salting with bcrypt in the admin registration route.
- Original logic allowed people to go back after logging out and view the dashboard. I added logic to prevent this.
- Tested formatting on mobile.

Challenges this week

- Understanding how to implement authentication and protected routes.
- Understanding how to use bcrypt for password hashing and salting.
- Handling some errors I encountered with the new features, such as ensuring the review form only accepts ratings between 1 and 5, and that the contact form submissions are properly saved and displayed on the admin dashboard.
- Implementing the admin features and ensuring that changes made in the admin dashboard properly update the marketing page content.
- CSS styling and layout for the new admin pages and popups. I had an idea of what I wanted, but I needed AI assistance to understand how to implement the design I had in mind.
- General learning curves with new concepts and technologies, such as authentication, bcrypt, and protected routes. The hw14b really helped me understand how to implement these features, though.

- I initially had a hard time understanding how the seed-admin.js logic worked.

Reflection

One challenge I had in integrating the passport.js logic was understanding how it works within the existing application. I was initially confused about how to structure the files and format the code and routes. Because we are using React and Express, I had difficulty conceptualizing how to set up the authentication logic across both the frontend and the backend. I also initially had trouble understanding how the seed-admin.js logic worked. The H14B assignment really set me up for success. I was able to use my HW14B file as a reference and template for how to structure my authentication logic. I also utilized AI tools to understand what changes I needed to make to my existing code to implement the new features. AI tools were very helpful this week, but I also feel like I came away with a better understanding of the concepts. I feel very positive about this week. I had a lot of fun developing the admin dashboard. I actually had to hold back on adding features because I had so many ideas for the admin dashboard and wanted to make sure I had time to implement the core features first. All in all, this week was a great learning experience.

Week 13 Updates

During week 13, the following features were implemented:
- Created the dashboard page using multiple elements: Dashboard.jsx, chart jsx files, component jsx files, data JSON files, dashboard CSS file, marketing page and dashboard page jsx files, arrivals js file, metrics js file, and origin.js file.
- Updated the file structure and code to support a multi-page application.
- Decided on the structure of the site and needed elements using PRD and wireframes.
- Selected which islands I wanted to pull data on.
- Retrieved, formatted, and inserted JSON data (Hawai'i Island, Maui, and Oahu) from DBEDT.
- Added an island selector (slctIsland)
- Added the Arrival Chart, Origin Chart, and Metric Cards elements using API calls and Chart.js.
- Decided on styling elements to be added to dashboard page: live stream, lava video, CTA, misc. strings
- Connected dashboard navigation from the site header.
- Updated the CSS files through multiple prompts as needed using GenAI.
- Researched volcanic data APIs and added VolcWidget. Saw the opportunity to have the background color change according to the aviation code, so I implemented that.
- Updated env: OpenWeatherMap API key in .env as VITE_WEATHER_KEY and tested API calls.
- Ran multiple tests on the app to ensure acceptance criteria are met.
- Ensured that widget updates automatically when the island selector changes.
- Added comments to the code as needed.
- Updated the readme with a guide for users.

Challenges this week

- Understanding how to set up the React project and structure the files.
- Understanding how to transition to a multi-page app.
- Resolving navigation button errors I encountered when paths changed.
- Understanding how to implement the weather widget.
- Understanding "react-router-dom"
- Fetching and formatting the correct data from DBDET.
- Deciding on CSS styling and formatting. Prompting GenAI was somewhat difficult as it struggled to render all of the elements in a pleasing way.
- CSS styling and layout. I wanted to do as much as I could manually, but if I wanted a complex and aesthetically pleasing design, I needed to use some AI assistance to understand how to implement the design I wanted.
- Understanding what makes the widgets update automatically.

Reflection

This week was another challenging week for me. Implementing a large number of changes took quite some time, but I am satisfied with the results, and I better understand what it takes to create a multi-page application. I especially appreciate the knowledge I gained on Chart.js visualizations. In the areas of code that were AI-generated/ assisted, I manually wrote comments in, and this very much helped me better understand the concepts involved. I struggled with the CSS styling a bit this week, as the AI-generated code was not reflecting what I had in mind. Eventually, I remapped my wireframe and was able to get to something satisfactory through prompting and manual adjustments. I also reformatted my marketing page to better align with my brand. All in all, I am satisfied with my dashboard.

Next week, I will implement admin authentication using Passport.js or JWT, create an admin login page, create a protected admin dashboard route, learn how to utlize bcrypt, use csurf middleware to implement CSRF protection on the login, decide on and develop my admin-only features, decide where my special offers and or announcements will post (admin-only feature), and design the layout of my admin page.

Week 12 Updates

During week 12, the following features were implemented:
- I went with option B to prepare for week 13: ensured the local Express server is running and implemented a fetch() call to /api/properties/:id using AI assistance to understand the needed syntax and structure.
- Added all the needed files and updated my database with the new fields for the marketing page.

- Added React functional components and JSX throughout.
- Each major section added as its own component file under src/components/: About, Amenities, Gallery, CTA, Hero, Header, Special Offer, Carousel, and Footer.
- Implemented responsive layout: readable on both desktop and mobile
- Tested routes in Postman. Verified property ID in MongoDB Compass matches the ID in the URL.
- Added alt attributes to all images and ensured color contrast meets WCAG 2.1 AA standards.
- Prompted AI to assist with API calls. Manually coded what I could using Keeper code, W3Schools, Udemy, and other linked React resources.
- Tested local express server and fetch() call in the browser. Verified data is being pulled from the backend and rendered on the marketing page throughout the coding process.
- Added const cors = require("cors"); to server.js and app.use(cors()); to allow cross-origin requests from the React frontend to the Express backend, as fetch was being blocked by CORS policy.
- Added more extensive marketing and property info to my schema and database to align with my PRD. Decided on tagline, amenities, and a package offering.
- Updated README to remove information that goes in the PRD.
- Manually added almost all comments to code (some using autofill from VS code) to increase my understanding and highlight what I manually coded (without AI assistance). Code I manually entered but used AI to help with structuring, syntax, or debugging is not labeled as manually coded, but I did make sure to understand all of the code I entered and the changes I made.
- Designed the wireframe and logo.
- Created property photos using some stock images and some GenAI.
- Manually coded some of my JSX files after I used AI to help me understand how to structure the components (using my local Express server) and what code to use. After the first few sections, I was able to understand the required structure and code and manually implement some sections without AI assistance.

Challenges this week

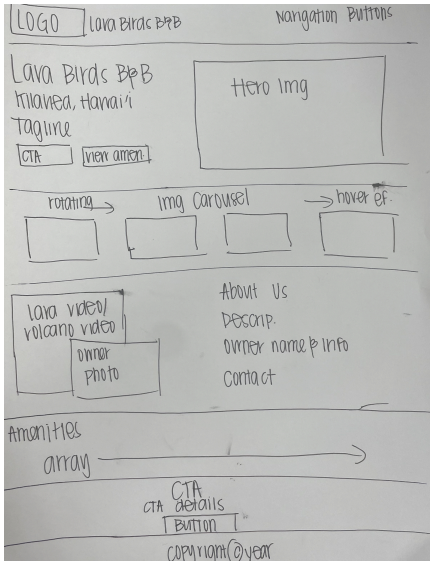
- Understanding how to set up the React project and structure the files.
- Understanding how to implement the needed changes without hardcoding, as I wanted to go with option B to prepare for week 13.
- Understanding what exactly needed to be done and what changes were feasible for me to make without AI assistance.
- Implementing the fetch() call to pull data from the Express backend and render it on the marketing page.
- Ensuring the marketing page is responsive and meets accessibility standards.
- Navigating the issues I encountered with the CORS policy and understanding how to resolve it by allowing cross-origin requests in the Express server.
- Troubleshooting MongoDB Compass to verify the property ID and ensure it matches the ID in the URL for the fetch() call.
- Figuring out how to add navigation links to my header.jsx file.

- CSS styling and layout. I wanted to do as much as I could manually, but if I wanted a complex and aesthetically pleasing design, I needed to use some AI assistance to understand how to implement the design I wanted.

Reflection

This week was very challenging but also very rewarding. I implemented all of the features above in my term project application. Although I did use AI tools, I began by creating all my files using Keeper code and Udemy as a reference to manually add what I could. As I went with option B, I needed some assistance with syntax. I decided to manually add/copy over all AI-generated code instead of prompting with Codex in VS Code for automated changes, and it really helped to increase my understanding of React. After a while, I began to better understand how to structure JSX files and was able to do some minor coding myself without any assistance. I also changed the information in my database to apply to the new one-property structure. I struggled a bit when it came to debugging and error handling, but I was able to use DevTools and AI assistance to figure out any errors I encountered. I did some back-and-forth prompting when it came to CSS styling, as I had a vision in mind that I could not feasibly accomplish without AI tools. I would write lengthy prompts, go over the code, copy the code over, run the program, and then start again. I am still not 100% satisfied with my styling and will probably change the styling a bit more next week as I add in my dashboard. Wireframing really helped me, so I will likely do the same next week. I also had fun making a Logo in Canva. I tried to find existing images online, but ended up using GenAI to create property images, which was more time consuming that I expected. It ended up being satisfying, though, as I could really tailor the images to apply to what was in my PRD. The PRD was also especially helpful this week as a planning tool.

Wireframe



Logo First Draft

Final Logos

Week 11 Updates

During week 11, the following features were implemented:
- Review schema (guestName, rating 1–5, comment, date) embedded in the Property document.
- Express routes: GET /properties, GET /properties/:id, POST /properties/:id/reviews.
- Mongoose query operators ($gte, $lte) added to filter properties by minimum/maximum rating.
- All routes tested in Postman.
- Postman collection exported and committed to the repo.
- Basic EJS template rendered at /properties listing all properties.
- Checked EJS rendering in browser.

Looking to week 12, the following features are ready to be implemented:
- React Marketing Page
  - Route is prepared in properties.js.
  - Endpoint returns all fields the marketing page will need: name, island, type, description, amenities, targetSegment, imageURL.
  - It will be important to meet the required sections, use React functional components, apply responsive CSS, and validate WCAG 2.1 AA compliance.
  - The features below are proposed for the marketing page.

Challenges this week

- Understanding how to structure the review schema and embed it in the Property document.

- Navigating Postman to test the new routes and ensure they were working as expected. Now understood.
- Understanding ($gte, $lte) operators and how to implement them for filtering by ratings